

Annah Singh

EE 472

11/19/24

Homework #8

```
1  module hw8 (output reg out,  
2      input wire clk, reset, in);  
3      reg [1:0] current_state;  
4      reg [1:0] next_state;  
5      parameter A = 0,  
6                B = 1,  
7                C = 2,  
8                D = 3;  
9  
10     always @ (posedge clk or posedge reset)  
11     begin: STATE_MEMORY  
12         if (reset)  
13             current_state <= A;  
14         else  
15             current_state <= next_state;  
16         end  
17     end  
18  
19     always @ (current_state or in)  
20     begin: NEXT_STATE_LOGIC  
21         case (current_state)  
22             A: if (in)  
23                 next_state = B;  
24                 else  
25                     next_state = A;  
26             B: if (in)  
27                 next_state = C;  
28                 else  
29                     next_state = A;  
30             C: if (!in)  
31                 next_state = D;  
32                 else  
33                     next_state = C;  
34             D: if (in)  
35                 next_state = B;  
36                 else  
37                     next_state = A;  
38         endcase  
39     end  
40  
41     always @ (current_state or in)  
42     begin: OUTPUT_LOGIC  
43         case (current_state)  
44             A : out = 0;  
45             B : out = 0;  
46             C : out = 0;  
47             D : if (in)  
48                 out = 1;  
49                 else  
50                     out = 0;  
51         endcase  
52     end  
53 endmodule  
54  
55
```

